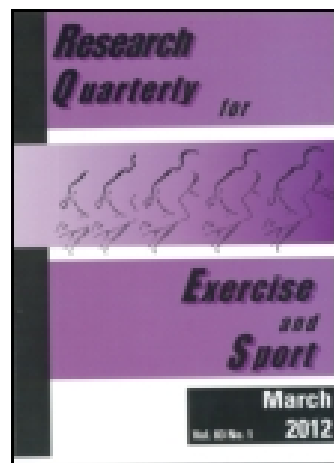


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On Becoming Skillful: Patterns and Constraints

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On Becoming Skillful: Patterns and Constraints

Jane E. Clark

Key words: motor skill, dynamical systems, locomotion, motor skill learning, development

How does someone become skillful? How did the great basketball star Michael Jordan or the renowned cellist YoYo Ma acquire the skills that make them great? For that matter, how did we learn the everyday skills of walking around without falling or the driving skills that enable us to venture out onto a busy interstate at rush hour? From the time of C. H. McCloy, for whom this lecture is named, understanding motor skill acquisition has been an important area of study. Today, far more is known about motor skills and becoming skillful than was known in McCloy's day. But while we know much more about the phenomenon, we have yet to formulate an adequate theory to explain motor skill acquisition.

In this paper, I maintain that the dynamical systems approach provides the theoretical framework from which such a theory can be derived. To make the case, I describe the dynamical systems approach and focus on two of its constructs, patterns, and constraints that hold particular significance for motor behavior. I follow this with a specific illustration of how this approach might be applied to the acquisition of locomotor skills. A central theme of the paper will be that skill acquisition is a *dynamic* and *developmental* phenomenon. An adequate theory on becoming skillful cannot be realized by limiting our focus to one period in the life span or to tasks that are not rich in context and complexity and real in their adaptive significance. Motor skill behavior changes over the life span, and it is that window that ultimately provides the view. To begin, I offer a reminder to those in the broad field of physical activity and physical education, in particular, that motor skillfulness is a crucial component in achieving a nation of physically active people.

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Preliminary Remarks on the Importance of Skillfulness

Movement is the way in which we act upon and interact with our environment. From the tasks of life to the sports arena and the art studio, motor skills form the mosaic of the human experience. Understanding how we move and how we become skillful is an essential for those in educational and therapeutic settings where the goal of motor skillfulness is an important objective in and of itself. But becoming skillful may also have a significant side benefit, namely engagement in physical activity. That is, those who engage in physical activity are more likely to be those who have acquired sufficient skill to do so. In other words, those who *can* play tennis, play tennis. Those who *can* swim, swim. Conversely, those with few motor skills (sports-specific or otherwise) may sit on the sidelines. The consequence of this inactivity is now recognized as a risk to maintaining health (U.S. Department of Health and Human Services Public Health Service, 1991).

Two years ago in another McCloy lecture, Blair (1993) concluded that there is substantial scientific evidence to suggest that inactivity increases the risk of dying from coronary artery disease and cancer. But today as we listen to the chorus of those who champion physical activity and fitness, let us not forget the importance of motoric skillfulness. This is not a new caution. As long as 50 years ago, McCloy (1940) warned of the insufficient attention given to skill development in the physical education curriculum. Today his concerns resonate for me as I watch the struggle over the agenda for a child's educational life, knowing well the time and effort it takes to become motorically skillful and the importance of the enterprise.

Becoming Skillful: The Problem

Becoming motorically skillful is a long process. We enter the world unable to feed, care, or transport our-

selves. Two decades later, we have built, expanded, and refined our skills to levels that not only permit us to survive but provide some with the expertise to perform such amazing locomotor feats as triple-toe loops on millimeter-wide ice skating blades. Indeed, for humans, the process of acquiring locomotor skills is no trivial matter. Most infants take their initial steps around their first birthday, having worked hard and long to find the solution to walking upright and independent. Of course that is just the beginning of bipedal locomotion. Running, galloping, hopping, skipping, and such culturally specific skills as the tango, the triple jump, and skiing are still before them. For movement scientists, these changes lead to questions of how this remarkable transformation occurs and what might be the fundamental principles that underlie skill acquisition. These are not easy questions to answer.

To act in or on the world, the mover must achieve and maintain a desired orientation to the task and/or the environment. In the case of walking, the desired orientation is *upright*. As a very experienced adult walker (think about how many hours of walking you have logged on your legs), the upright posture is easily achieved. But for the novice walker, who began sitting independently only 6 or 7 months earlier, maintaining the upright orientation over a two-footed base is a challenge made even more daunting by an everchanging support base (i.e., from one foot to the other). The problem becomes more serious as the infant attempts to walk in different and complex environments. It should not be surprising, therefore, to see newly walking infants fall over toys in a cluttered room or bump into other toddlers as they play. If we view the problem from the perspective of what the central nervous system (CNS) must control and coordinate, the task is all the more overwhelming. Consider the control problem for a system that has over 100 billion neurons, a body with over 1,000 muscles and at least 100 times more motor units. And the solution desired from all this complexity is a movement that is automatic, efficient, and adaptive.

How the CNS achieves this task has been viewed historically from two perspectives or paradigms: neuromaturation and information processing. The neuromaturation perspective has been used primarily for skill acquisition in infants and young children, whereas the information-processing perspective has been the preferred interpretive framework for adult motor skill acquisition.

Neuromaturation Perspective

Clearly, becoming an independent walker presents the infant with a host of problems. How the child achieves a solution to these problems is often framed in

a neuromaturation perspective (Forssberg, 1985; McGraw, 1963). This perspective sees the unfolding motor sequences of infancy as a direct result of nervous system maturation. Presumably the brain structures grow and behavioral change results. An infant's parents watch in wonder as motor skills appear almost magically with ostensibly little adult intervention (i.e., teaching). No doubt skills such as walking require far less environmental support than other later appearing skills such as riding a bike. However, walking, like all motor skills, results from learning—not from some magical maturational process that gives us motor skills as birthday presents. Interestingly, motor skills acquired by adults are rarely framed in maturational terms. So we must invest our scientific explanations of infant motor development in maturational tautologies. If one carefully watches an infant struggling to take that first walking step, the similarities with an adult challenged by a new sport skill are striking. Just because you did not “teach” the infant to walk does not mean the infant did not “learn” to walk. Likewise, the adult's behavior is not solely dependent on environmental factors. Just as the infant's maturational status is a component of the resulting behavior, so too must biological substrates play a role in adult motor skill acquisition and performance. Motor skill acquisition is the process of learning motor skills in the context of the performer's previous experience and biological status. Neither component has a privileged importance.

Information-Processing Perspective

For the last 30 years, our explanations of adult skill performance and learning have, with one exception (i.e., Adams', 1971, closed-loop theory), been couched in an information-processing framework (cf. Fitts & Posner, 1967; Schmidt, 1975). Using the computer as a model, the information-processing approach has provided a rich and powerful explanatory scheme. Computers store, retrieve, process, and sometimes lose information. Likewise, motor skills are stored, retrieved, and lost. Once you have written a computer program, it is not difficult to imagine that movement can be controlled by a “motor program.” Metaphors abound. But although these comparisons are compelling, the metaphors are insufficient for explaining motor skill performance and learning. In brief, the debate falls along two lines of dispute. (For a full discussion of the arguments, the interested reader is referred to Kelso, 1981; Saltzman & Kelso, 1983; and Turvey, Fitch, & Tuller, 1982.) The first centers on the assumption of the information-processing perspective that movements arise from plans and instructions commanded by an executive. In this view, movements are prescribed by this executive in much the same way as a pianist might play a musical score.

But who is this executive? Is there some little man in our head who oversees our motor programs? Where do motor programs come from? The second line of arguments lies with the issue of complexity. Both the biological system itself and the environment within which it finds itself are complex. Information-processing approaches work well for simple behaviors, such as those found in reaction time tasks, but when forced to explain the versatility of a skilled basketball player, the nonmuscular forces arising in human motion, or the appearance of a new movement, they fall short.

In the last 10 years, an alternative to the neuro-maturational and information-processing approaches has emerged. This alternative perspective, the dynamical systems approach, sees movement not as a prescription or a set of instructions issued by some little man in our head or imposed by our genes, but rather as an emergent phenomenon that arises from the constituent elements of a complex dynamic system.

Dynamical Systems Perspective

The general field of dynamical systems, perhaps better known under the terms chaos, complexity, or nonlinear dynamics, originated in the mathematical and physical sciences to explain systems that change over time. The inception of dynamical systems dates back almost a century, but it is only in the last decade that the approach has expanded widely to include such diverse applications as meteorology, economics, and neurophysiology (e.g., Glass & Mackey, 1988; Haken, 1985; Kelso, Mandell, & Shlesinger, 1988; Morrison, 1991). In the movement sciences, the approach was first introduced in 1980 by Kugler, Kelso, and Turvey. In their seminal paper, Kugler et al. proposed a view of movement that would set the course for a totally new conception of motor skill performance and acquisition. Much has been written since that first paper, and according to most of the criteria defined by Kuhn (1962), a new paradigm has evolved (see Abernethy & Sparrow, 1992, for a full discussion of the paradigm issues).

Although a comprehensive discussion of the dynamical systems approach is beyond the scope of this paper, a brief description of four essential concepts that sit at the center of the approach, as it is applied to motor skills, is presented. These concepts are constraints, self-organization, patterns, and stability.

Constraints

The first concept that is critical to a dynamical systems approach to motor skills is the idea that motor behavior arises from a system that is surrounded by constraints. These constraints limit or set the boundaries

for the system. Newell (1986) categorized these constraints into those found in (a) the organism, (b) the environment, and (c) the task at hand. The organism constraints refer to those properties embodied in the person. These would include not only the physical characteristics such as height, weight, and body shape, but would also include the cognitive, psychological, and emotional attributes of the individual. The structure of the knee and the attentional state of the performer are examples of constraints that might shape or limit a movement. The environmental constraints arise in both the physical and sociocultural environments of the mover. Walking on the moon or walking across the stage at a graduation ceremony reveal different types of environments that constrain movement. Culture also acts as environmental constraints that shape movements. Although these constraints may be more subtle than the physical ones, they are nonetheless ever-present surrounds to the actor. Reed (1993) referred to these sociocultural constraints as the *field of promoted action*; that is, those actions that a culture promotes and nurtures. But it is the task of here and now that offers the final shape to a movement. Against the backdrop of the organism and environmental constraints, it is the requirements of the task at hand that ultimately forms the movement that we see. Picking up a cup from which to drink gives a reaching movement different form than picking up an apple to eat. Task constraints marshal the system into the behavior we observe. From a dynamical systems perspective, movements emerge from constraints.

Self-Organization

Constraints alone are not sufficient to give us movement. There must be some process that gives rise to organized action. In dynamical systems, this process is self-organization. Highly complex systems comprised of multiple simpler elements governed by dynamical principles will spontaneously develop organization or patterns; that is, they will self-organize (Werbos, 1994). Systems of high dimensions, such as the neuromusculoskeletal system, will form patterned action from the constituent elements. For example, energetic systems that are far from a state of equilibrium will find a stable equilibrium point, and therefore an organization, under certain sets of constraints. No little man in the brain is needed to command the system to organize in a specific way. Rather, a pattern emerges from the constraints through the process of self-organization.

Patterns

Patterns and pattern formation are fundamental to the dynamical systems approach. From an enormously complex system of billions of neurons and thousands of motor units, a low dimensional description—a pat-

tern—emerges. These patterns result from a dynamical systems' collective dynamics. We can represent these patterns geometrically in a phase portrait. This is done by creating a geometric model for all possible states of the dynamical system. This is referred to as the state space (Abraham, Abraham, & Shaw, 1992; Abraham & Shaw, 1982). Attractors are formed when the system is drawn to specific regions of the state space. An example of a phase portrait for the motion of the lower leg (i.e., the shank) is depicted in Figure 1. The state space is defined by the angular inclination of the position of the shank on the x axis and the shank's angular velocity on the y axis. The space defines all the possible states of the walking shank. The periodic trajectories mapped in the state space represent the attractor formed by this system's dynamics. This attractor compresses the behavior of multiple muscles of the leg, as well as the motor and sensory neurons and motor units, into one dynamic pattern.

Stability

Attractors represent stable collective states of a dynamical system. Small perturbations, such as stepping on a pebble, would not dislodge the system from this region of the state space. Rather, the system once perturbed would return quickly to the original trajectory. Stable patterns are very low in variability. When variability increases, the system is unstable. The loss of stability is characteristic of a system in transition. When one stable behavior, such as crawling, gives way to another stable behavior, such as upright walking, our predictions would be that in the transition, there would be increased variability. Further, because behaviors emerge from constraints, it follows that changes in a pattern can be attributed to changes in one or more constraints. When scaling a constraint changes a system, that constraint is referred to as a *control parameter*.

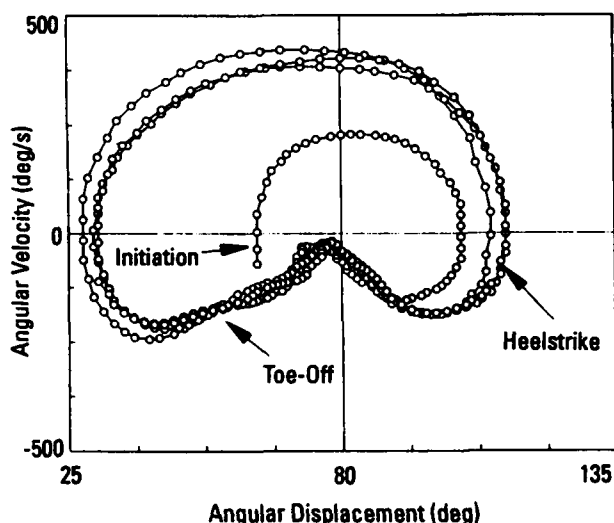


Figure 1. Shank phase portrait of four adult walking cycles.

Motor Skill Acquisition From a Dynamical Systems Perspective

The dynamical systems approach to motor skill acquisition offers a potentially rich framework within which to form a principled account of how one becomes skillful (e.g., Zanone & Kelso, 1994; Thelen & Smith, 1994). Here I focus on the predictions that derive from this approach relative to the transition into *new* behaviors. The acquisition of independent walking is the exemplar for this discussion. Although infants take their first independent steps around 1 year of age, the entire 1st year of life is spent assembling the necessary components (i.e., organism constraints) for this action. Thelen (1986) offered the hypothesis that there might be eight potential components (subsystems) that must come to some critical value (i.e., level of maturity) before walking will emerge. These eight components are pattern generation, articular differentiation, postural control, visual flow sensitivity, tonus control, extensor strength, body constraints, and motivation. If one or more of these components (i.e., constraints) has not reached some critical level, then it will act as a rate limiter holding back the emergence of the new behavior. Thelen's own work (Thelen, Bradshaw, & Ward, 1981; Thelen, Fisher, & Ridley-Johnson, 1984; Thelen & Ulrich, 1991) and earlier work by McGraw (1932) clearly show that pattern generation in the form of the stepping reflex, kicking, and treadmill walking is in place very early in the life of the infant. The time course of the other components remains an open question, as well as whether these are the relevant constraints (or rate limiters) for walking.

In the section that follows, I illustrate how the dynamical systems approach can be used to study skill acquisition with work I have been conducting over the last decade with my colleague, Sally Phillips, and a number of graduate students. For the most part, we have followed the research strategy outlined by Kelso and Schöner (1988) in which we first seek a collective variable (a low dimensional description of the system) or variables from which an attractor can be identified. The attractor (i.e., the pattern) is then examined for stability and instability, and potential control parameters are sought. Our task then is to describe our dynamical system (i.e., walking) as a pattern and then identify those constraints that when scaled will influence the pattern.

Learning to Walk

One of the most striking feats of infancy is the achievement of independent, upright, bipedal locomotion (i.e., walking). Arriving in the world totally dependent on caregivers for moving from one place to an-

other, the human infant gradually, over the 1st year of life, learns to roll over, scoot, crawl, pull to a stand, and cruise along the couch. Somewhere around the infant's first birthday, those first independent walking steps are taken. It is from that point and thereafter for the next 6 months of practice that we focus our analysis on learning to walk.

Our first research goal was to find a collective variable that would capture walking as a global pattern that could then be studied for its stability and instability. In gait, finding such a variable is no trivial matter. The literature is replete with variables to describe one or more aspects of walking. For example, Sutherland, Olshen, Cooper, and Woo (1980) reported time series analysis on 12 measures of body position, 6 force measures, 7 muscle activation sequences, and 9 time/distance measures for one cycle of walking. As walking experience increased, Sutherland et al. found that all the measures improved. So how are we to find a single variable that would capture the complex multidimensional system as it learns to walk? First, we limit our focus to explaining the development of *coordination* in walking. Rather than looking for all the possible ways in which walking changes, we sought a narrower scope. In particular, we began with the coordination between the two legs, the *interlimb* coordination. After all, to walk, the two legs must work together in supporting and propelling the body to the intended location. We then asked the question: How is it that the legs come to be coordinated with each other as they achieve the task of walking upright and independent?

Interlimb Coordination

The collective variable we selected for capturing the interlimb coordination for walking was the phasing relationship between the legs as defined by footfalls. A gait cycle within one leg is defined from one event (e.g., footfall) to the next appearance of that same event. If we look at the coordination between the two legs, we could ask what is the relationship between the legs as defined by specific events in each one's cycle (e.g., footfalls)? In the case of walking, the footfall of one leg falls halfway between the footfalls of the other leg. In other words, the legs are out of phase with each other by 180° or, expressed as a percentage, one leg starts its cycle halfway (i.e., 50%) through the other leg's cycle. The interlimb coordination for adult walkers, expressed as a percentage, is on average a 50% phasing relationship (Clark, Whitall, & Phillips, 1988).

When first learning to walk, how will infants coordinate the two legs (i.e., their temporal phasing relationship) and how long will it take them to find the adult-like coordination? The answer is surprising. As it turns out, infants at the very onset of independent walking have an adult-like mean phasing of 50% (Clark et al., 1988). But the interesting aspect of this data is not that

the infants walk on average at a 50% coordinative relationship. Indeed, according to Raibert (1986), it may be that this alternating, symmetric coordination is actually the easiest solution for a two-legged walker to adopt. What is interesting and also what we would predict from a dynamical systems perspective are the findings on the variability of the temporal phasing relationship. In Figure 2, we see the mean variability within an infant across walking experience. This average is calculated from the standard deviation about each infant's mean temporal phasing. Clearly, the coordinative relationship is unstable in the new walkers. While they attain a 50% mean phasing, they are quite variable around that mean. With increasing practice, however, the relationship becomes more stable (i.e., less variable). After 3 months of walking experience, this variability is not significantly different from that of the adult walkers (Clark et al., 1988).

Another finding from this study that is of interest is the effect of supporting the new walker. When our newly walking infants' hands were held by the experimenter (see Figure 2, N-Sup), interlimb coordination became more stable, with the variability similar to that seen in the 1-month walkers. This was our first demonstration that *posture* might be a control parameter (or rate limiter because it was holding back development) for learning to walk. In later studies, we would find similar clues pointing to this global parameter.

In our next experiment on interlimb coordination, we sought to explore more fully the stability of the collective variable. To do this, we conducted a perturbation study (Whitall, Block, & Clark, 1992). In dynamical systems, one way to test whether a system is stable or not

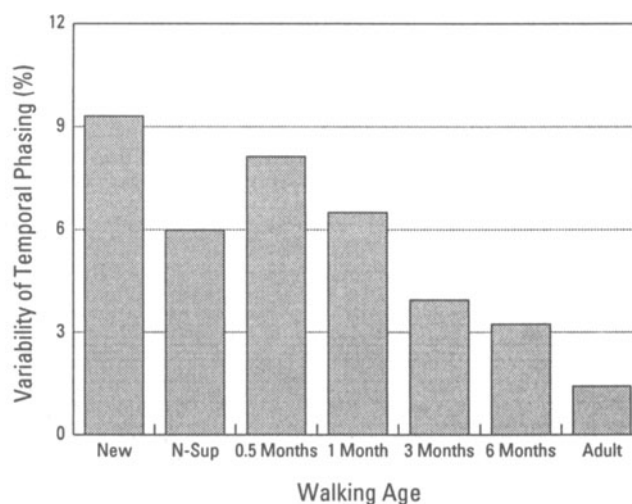


Figure 2. Variability of temporal phasing for interlimb coordination at different walking ages. N-Sup = new walkers with their hands held by the experimenter. Redrawn from "Human Interlimb Coordination: The First 6 Months of Independent Walking," by J. E. Clark, J. Whitall, and S. J. Phillips, 1988, *Developmental Psychobiology*, 21, p. 451.

is to poke at it; that is, to perturb it. If a system is stable, it will resist the perturbation; if not, the perturbation will increase its instability and may be forced into a new behavioral pattern. Because we had found that new walkers immediately demonstrated a 50% phasing relationship, although more variable than more experienced walkers, we wanted to know more about the stability of the coordinative relationship. As a system in transition, could we push the infant out of the 50% phasing? To test this hypothesis, we affixed a small weight (5% of the infant's body weight) to the ankle of one leg. On a 20-lb infant, which was the average weight of infants in our study when they first began to walk, this amounted to a 1-lb weight being taped to the ankle. To control for adding weight per se to a young walker, we also had the infants walk with the same weight taped snugly to their lower backs.

Our first finding was rather robust. When we put the weight on the new walkers—that is, those that could walk only three or four steps before falling—our novices refused to walk. Several just stood there looking down at their legs. Some just sat down and crawled away. One or two tried to walk, but without success. Clearly, the newly walking infant was perturbed by the weight. Two weeks later, all the infants were willing to attempt a step or two with the weights. But it was not until 1 month of walking experience that the infants walked the same number of steps with and without the weight. Temporal phasing for the 1-month walker and then at 3 and 6 months of walking revealed again that the 50% phasing relationship was on average the attractive pattern of coordination whether walking with a weight on the ankle or on the back or walking without a weight. No differences were found in temporal phasing variability, suggesting that the weight did not make the interlimb coordination less stable.

Thus, after 1 month of walking experience, our young novice walkers had found an attractive, stable pattern of interlimb coordination. Although it took them another 2 months of walking experience to demonstrate the same low variability as the adult walkers, the weight perturbation revealed that the 1-month walkers had found a pattern of interlimb coordination that resisted this type of change. Within a few months of their first attempts at independent walking, the infants had found a solution to get from here to there—hands-free, bipedally, and with reasonable stability. But although we had found at least one variable that described the coordination of walking, we had not identified an attractor to describe the behavior. In addition, we had addressed only one type of coordination; that is, the coordination between the legs. It was clear that this was only one of the coordinative tasks of walking. For example, each leg is comprised of over 50 muscles and “breaks” at the ankle, knee, and hip. How is it that the newly walking infant coordinates the muscles within

each leg? We thus turned to examining the *intralimb* coordination of the lower extremities.

Intralimb Coordination

The work on interlimb coordination led us to investigating a dynamic model of walking in which the legs were viewed as coupled, nonlinear pendula. Such a dynamic model would have considerable promise for going beyond description to explanation in understanding skill acquisition and might give us a description of the system as an attractor. A simpler but similar model had been suggested some time ago in a paper by Kelso, Holt, Rubin, and Kugler (1981) in which they demonstrated that coordination between the right and left forefingers displayed properties of coupled, nonlinear, limit-cycle oscillators. In the case of walking, motion of the two major segments of the leg (i.e., the thigh and shank) swing to and fro much like pendula coupled at the knee and fixed at the hip. Unlike freely swinging pendula, however, the leg segments come in contact with the ground. And for a major portion of the gait cycle, the two segments act as an inverted pendulum as the body rides over the foot through stance.

To investigate whether our walking legs act as coupled limit-cycle systems, we began by mapping the action of the segments in state space. As described earlier, the state space is a geometric space within which our dynamical system (i.e., the walking leg) can be described. Figure 1 is a state space description of the shank's action during four cycles of walking as defined by the state variables of angular position and velocity of the segment (for a description of the thigh, see Clark, Truly, & Phillips, 1993). Position and velocity are the two state variables that capture the system's behavior. Each of the circles (O) on the graph in Figure 1 represents the state of the system at any one time. The inner trajectory marks the initiation of the walking cycle. The trajectory moves clockwise in state space and within one cycle has found the region of the space where all the other cycles are attracted. The brief period from initiation until the trajectory settles into a periodic orbit is called the *transient* and demonstrates why the picture that emerges is called an *attractor*. Mapped in state space, our walking shank's behavior forms a periodic, closed orbit. The indentation between heelstrike and toe-off represents the dissipation of energy as the shank contacts the ground and the reinfusion of energy as the shank pushes off the surface at toe-off. This type of energy exchange is characteristic of a limit-cycle system.

Returning to our skill acquisition story, then, we consider our newly walking infant's segmental behavior for evidence of a similar limit-cycle attractor. Based on our predictions about dynamical systems in transition, we might expect that the infant would find a solution in

much the same way as they had done in their interlimb coordination. That is, an adult-like solution would be found, but it would be unstable. Indeed that was the case. In Figure 3a, an exemplar for one of our newly walking infants is presented (see Clark & Phillips, 1993, for evidence of three infants followed longitudinally for 1 year). Here we see that the infant's shank trajectory is periodic and closed (i.e., limit cycle-like), but the behavior is quite variable. Unlike the adult shank trajectory where one orbit fell upon another, the infant's trajectories (except during stance) are not attracted to the same region of the state space. However, after 4 weeks of walking experience (see Figure 3b), the infant has settled into an attractive regime. Although the shank's trajectory is still not quite the same in its shape as the adults' trajectory during stance (the lower portion), it is evident that an attractive limit-cycle behavior has formed. So in a similar fashion to the findings for interlimb phasing, we see that at the onset of walking, the system takes on an adult-like pattern, but one that is quite variable.

Although the behavior of a single segment, such as the shank or the thigh, is revealing, it does not directly address coordination within the leg. To examine the intralimb coordination, we need a measure of the relationship between the shank's trajectory in state space and the thigh's trajectory (or the foot). To calculate this measure, we first transformed the Cartesian ($x, \dot{x}$) to polar coordinates, with a phase angle $\phi = \tan^{-1}[\dot{x}/x]$ and radius. In our convention, a ϕ of 0° represents a point on the trajectory where the angular velocity is 0 deg/s and the segment is rotated forward as far as it will move in that cycle

(at a time after toe-off, in a position where the hip is ahead of the foot). Conversely, a ϕ of 180° is also at 0 deg/s , but the segment is now back as far as it will rotate (near to heelstrike, where the foot is ahead of the hip). Starting at toe-off, we can calculate the relationship between the shank and the thigh at every point in the walking cycle. Across the walking cycle, this phasing relationship in an adult would look like the time series displayed in Figure 4a. Unlike interlimb coordination, where we examined only one point in the cycle (i.e., footfall), with intralimb coordination we see the coordinative relationship throughout the walking cycle. Now rather than a fixed point that is 180° out of phase, we see a relationship that is nonlinear and changing throughout the cycle. But while it changes within the cycle, the time series pattern is remarkably similar over individuals walking at their preferred speeds (Clark & Phillips, 1993).

Figure 4a also includes the time series for our new walker. Here we see for the first time a pattern that is not adult-like at the onset of the new behavior. The shank-thigh coordination is quite different in the new walker's first half of the stance phase (from about 25–65% of the walking cycle). So, too, the late portion of swing (15–25%) differs from the adult. In the adult walker, the phasing relationship from toe-off (0% of the walking cycle) to heelstrike ($\sim 40\%$ of the walking cycle) begins at about -80° , with the thigh moving through state space ahead of the shank. (Note that in our measurement convention, a negative phasing relationship means the thigh is leading the shank in its state space trajectory.) Then in late swing (~ 20 to 40%), the thigh stops its motion and the shank rapidly overtakes it. The

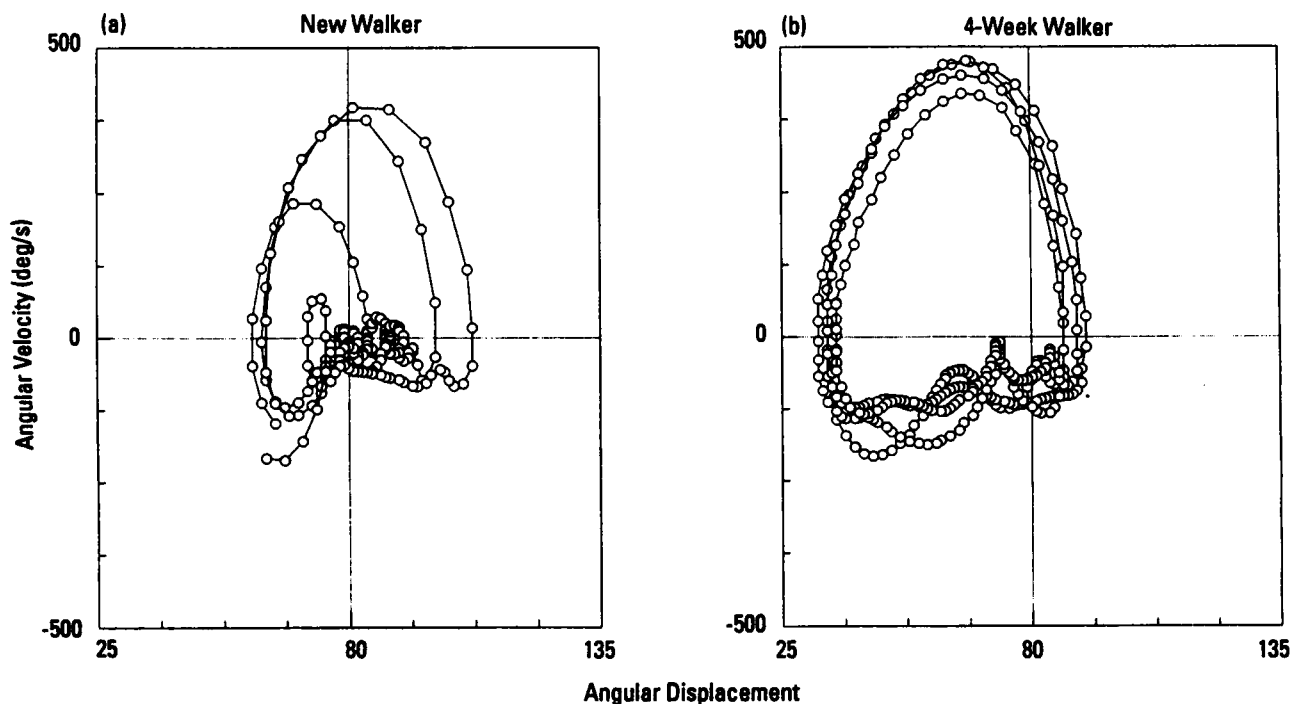


Figure 3. Shank phase portrait for an infant at the onset of walking and 4 weeks later.

new walker, in contrast, has a much shorter swing phase (25% compared to the adult's 40%) in which the thigh stays ahead of the shank throughout.

Two months later (see Figure 4b), however, our young walker has found the same coordinative relationship as the adult, although lower in magnitude. In our longitudinal study of infant walkers (Clark & Phillips, 1993), we concluded that by 3 months of walking infants had found the adult-like shank-thigh relationship. We also found a very interesting effect when we perturbed the young walkers with the ankle weight. One-month walkers without any perturbation have an intralimb coordination similar to the new walker (see Figure 5). With the ankle weight, the relationship between the thigh and the shank becomes more adult-like in late swing (between 20 and 30%). The weight did not destabilize the infant, but rather seemed to elicit a more mature pattern of coordination. Indeed, the infant seemed to exploit the weighted shank so as to move it faster than the thigh—a coordinative relationship seen in the more experienced walkers.

What might this perturbation tell us about how the infant learns an adult-like intralimb coordination? Although speculative, it might be argued that the weighted ankle promoted a motion that had not yet been found by the young walker, namely a freely swinging shank off a fixed thigh. What the infant learns with walking experience is how to exploit the swing limb dynamics such that the shank gets much of its motion for

free (i.e., without muscular input).

Our longitudinal study (Clark & Phillips, 1993) also found indirect evidence that posture is a critical rate limiter to mature walking. In a walking cycle, periods in which both feet are on the support surface (referred to as double support) are interspersed with periods in which only one leg supports the body (referred to as single support). When new walkers first attempt independent steps, they spend about 60% of their walking cycle in double support in contrast with adults who spend about 20% of the cycle with both feet on the surface. After 2–3 months of walking experience, infants give up their land-bound strategy and walk with about the same amount of time spent in double support as the adults. This, of course, is about the same time that we see adult-like stable behavior in both intralimb and interlimb coordination. Although posture was not manipulated, the evidence again points to the importance of this system in the development of upright, bipedal locomotion.

On Becoming Skillful

Although briefly presented, our work on how infants come to coordinate their legs as they learn to walk upright and independent offers several findings that illustrate the principles of a dynamical systems approach

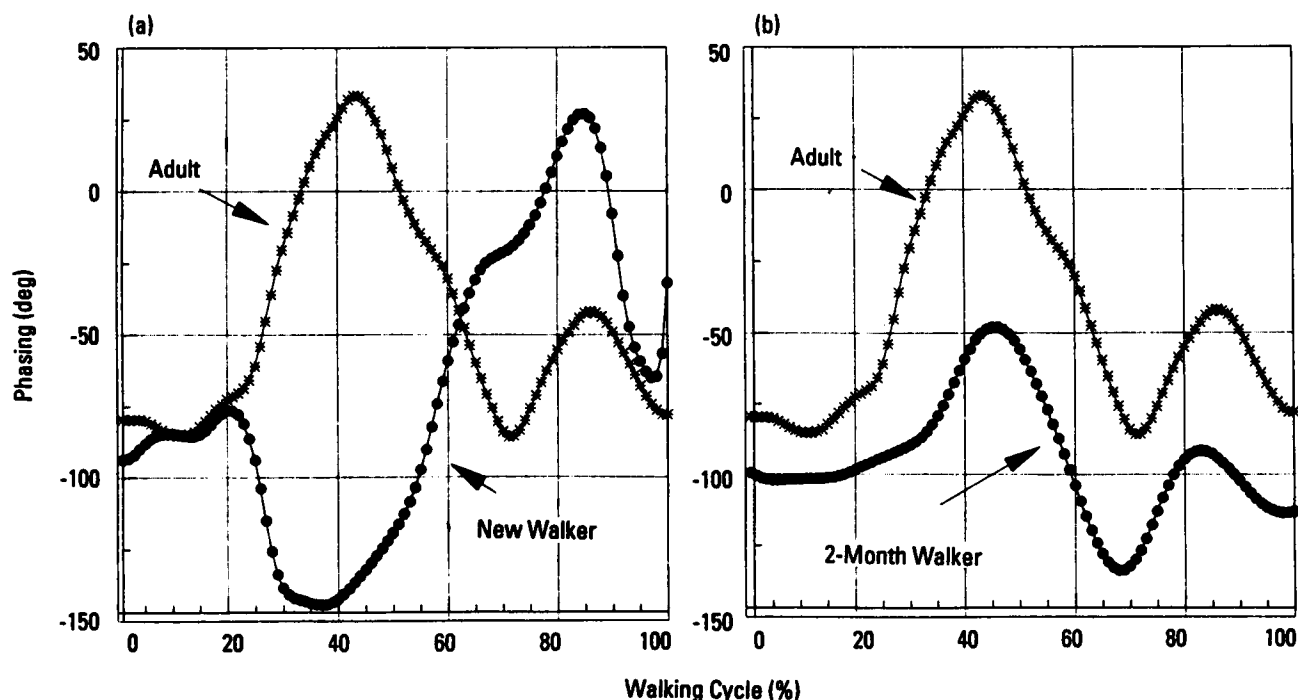


Figure 4. (a) Relative phasing between the shank and thigh segments for an adult and a new walker and (b) the same measure after the infant had been walking 2 months.

to motor skill acquisition. The first of these principles is that behaviors emerge from the constraints that surround them. When a behavior first appears in the infant's repertoire, that behavior emerges from the organism, environment, and task constraints of the moment. Another way to frame this principle is to examine the effect that changing constraints have on behavior. In our work, we can demonstrate one clear piece of evidence on this point, namely the findings on interlimb coordination when the infants were supported. Providing postural support stabilized the phasing relationship between the legs such that our new walkers looked like they had been walking for about a month. Clearly posture per se is not a satisfactory explanation. But this and our findings on changes in time spent in double support offer a clue that posture may well be an important area for more in-depth probe into just what it means to say posture is a rate limiter for walking.

A second principle on skill acquisition that emerges from the dynamical systems perspective is that dynamic patterns are the appropriate units of analysis. Indeed, the process of learning a skill may be conceptualized as the process of finding an attractive dynamic state that meets the task demands. Many attractive states are possible. For example, crawling is an attractive dynamic state, as is standing or cruising along the couch. But to walk, the infant must find the constraints that will hold together a hands-free, bipedal, propulsive state. Much of what the infant does in those months before achieving an independent walk is to explore the dynamic landscape in which walking is one of many attractive states.

The third principle that is illustrated in the story on learning to walk is that behaviors in transition are unstable. Although motor learning theories have classically seen variability as a nuisance or as an error, within

a dynamical systems perspective, variability is not only expected but it is a critical property of the system. Stable patterns represent behavioral states that are reproducible and independent of each other. As the system passes from one state to another, there is a period of instability (Schöner & Kelso, 1988). Variability then points to times when the system is in a state of transition. It also reveals a time when the potential relevance of a constraint as a control parameter can be explored. But most important, the dynamical systems perspective highlights the importance of a system that is not constant nor fixed, but rather dynamic, variable, and adaptive. To learn, a stable state must give way to a new state. Crawling must give way so that the infant can find that region of the dynamic landscape where walking is the attractive and ultimately stable behavior.

Finally, I would reiterate that although skill acquisition is a dynamic process, it is also a developmental process. Motor skill acquisition takes place across the life span. To say that our work on how infants acquire walking is motor development while work on adults learning a badminton serve is motor learning is to make an artificial distinction that limits our understanding of how one becomes skillful. There are many windows on the system. Here I would argue that the onset of walking is one window that offers a unique view on skill learning. Becoming skillful is a dynamic and developmental process—everchanging and present throughout the entire life span.

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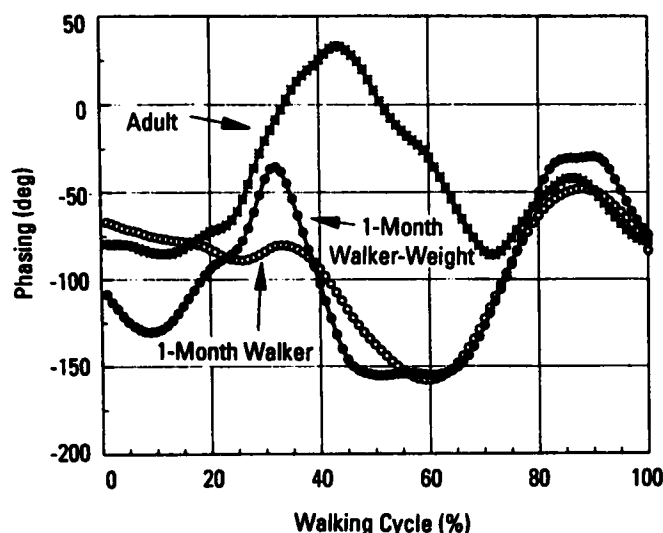


Figure 5. Relative phasing between the shank and thigh segments for an adult and a 1-month walker with and without an ankle weight.

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